

Achilles Tendinopathy: Evaluation, Rehabilitation, and Prevention

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Abstract

Achilles tendinopathy is a common overuse condition that is characterized by degenerative, cumulative tissue microtrauma. It is largely a clinical diagnosis in which the patient typically presents with localized pain that is worse with tendon-loading activities. Imaging modalities may be helpful with the diagnosis of difficult cases or in the planning of interventional procedures. Midportion (noninsertional) and insertional tendinopathy have distinct features and differences for therapeutic paradigms. Overall, Achilles tendinopathy has a good clinical prognosis with most patients improving with activity modification and rehabilitation, with a focus on progressive tendon-loading. Recalcitrant cases may require adjuvant treatment with procedures (e.g., injections, shockwave therapy) and rarely surgical intervention.

Introduction

Achilles tendinopathy is a common ankle condition in both the general population and athletes (1,2). It is among the most common musculoskeletal injuries seen in running sports, with an incidence ranging from 9.1% to 10.9% in runners and a lifetime risk exceeding 50% in former elite male distance runners (1,2). Achilles tendinopathy is a degenerative process caused by overuse and cumulative microtrauma with both intrinsic and extrinsic risk factors (3,4).

The typical clinical presentation of Achilles tendinopathy includes a triad of pain, swelling, and impaired function, especially with walking and impact exercise (2). Intrinsic risk factors include patient demographics, such as older age, male sex, poor vascularity, hyperpronation and other biomechanical factors, and systemic diseases including metabolic and rheumatologic disorders (2,4). Extrinsic risk factors include overuse, training errors (rapid increase in impact exercise volume), and certain medications that may predispose to tendon injury

including steroidal medication and the fluoroquinolone class of antibiotics (4).

Achilles tendinopathy can be divided into two distinct categories based on location: insertional tendinopathy and midportion (noninsertional) tendinopathy, which have unique features and differences in treatment strategies (5,6). Midportion Achilles tendinopathy, located 2 cm to 6 cm proximal to the insertion, is more common, representing up to 66% of cases (4). In contrast, insertional Achilles tendinopathy occurs at the insertion site on the calcaneus. Bone spurs, enthesophytes, and calcifications near

the insertion may be observed and can contribute to retrocalcaneal or superficial calcaneal bursopathy (4,5).

Management is typically nonoperative with options, including relative rest and activity modification, physical therapy (PT) with evidence-driven rehabilitation, and nonoperative adjuvant therapies. Generally, a good prognosis is expected for most patients with Achilles tendinopathy (6). However, this condition can be challenging to treat in a subgroup of patients and can lead to significant morbidity. The aim of this narrative review is to present the best-practice approach to clinical evaluation of Achilles tendinopathy and to outline the current evidence for rehabilitation and therapeutic treatment strategies for both insertional and midportion Achilles tendinopathy.

Methods

A literature review was conducted using PubMed/MEDLINE and Google Scholar electronic databases from inception up to January 2021 with combinations of the following relevant keywords/MeSH terms: Achilles tendinopathy, insertional, mid-substance, midportion, noninsertional, treatment, rehabilitation, PT, prevention, nitroglycerin patches, corticosteroid, orthobiologics, platelet-rich plasma (PRP), tenotomy, extracorporeal shockwave therapy (ESWT), surgery. The search results were screened for studies relevant to the topic and only English-language studies were considered for inclusion.

Anatomy and Pathophysiology

The Achilles tendon is the largest and strongest tendon in the human body, conjoining the medial and lateral gastrocnemius

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muscles and the deeper soleus to its insertion distally on the posterior surface of the calcaneus, with some fibers adjoining the plantar aponeurosis (7). In 6% to 8% of patients, the plantaris contributes to the Achilles tendon medially (7). The primary function of the tendon is ankle plantar flexion. It also aids in stabilization of the tibiotalar joint during gait. The retrocalcaneal bursa lies deep to the tendon, adjacent to Kager's fat pad within the retrocalcaneal recess at the posterosuperior calcaneus. The superficial calcaneal bursa is located between the distal tendon insertion and the overlying skin.

Achilles tendon architecture is predominantly composed of well-organized tenocytes and tenoblasts dispersed within a hierarchy of collagen (primarily type I) fibrils, oriented longitudinally in a network of extracellular matrix that creates a robust, thick, and round tendon that is able to transmit significant force (8). The tendon, innervated by the sural nerve, lacks a true synovial sheath and is instead surrounded by an elastic, connective tissue layer called the paratenon. While the majority of the tendon itself has poor vascularity, the paratenon supplies vasculature to the anterior portion of the tendon greater than the middle and posterior portions (2,8,9). The midportion of the tendon is the least vascularized region with a watershed zone 2 cm to 6 cm proximal to the insertion (8,9). This poor vascularity contributes to difficulty in self-repair following microtrauma.

Heterogeneity exists in the pathogenesis of Achilles tendinopathy. The tendon disease state includes a primarily disorganized collagen construct histologically with a paucity of inflammatory cells (4,9). The thickened, pathologic tendon demonstrates an increase in irregularly shaped tenocytes and an increased expression of type III collagen and glycosaminoglycans, as well as abnormal neovascularization (1,8,10). An increase in mechanoreceptors and pain receptors has been found within the neovascular walls, thought to play

a role in pain generation (11). Bursitis may coexist with Achilles tendinopathy and contribute to pain (4,5).

Clinical Evaluation

Presentation and Physical Examination

Patients with Achilles tendinopathy typically present with pain located at the midportion of the tendon or at the insertion on the calcaneus. The Table highlights key clinical and imaging characteristics and treatment differences between midportion and insertional Achilles tendinopathy. Patients classically describe focal tenderness, localized tendon thickening, and functional impairment with load-bearing activities. Patients often experience morning pain or stiffness during the first steps of the day or pain at the beginning of an exercise session. Pain may subside with continued activity. However, in chronic cases, pain may be persistent and occur at rest.

A broad differential of posterior ankle/hindfoot pain should be considered. Bony causes of posterior ankle pain include posterior ankle impingement, os trigonum syndrome, talar osteochondral lesions or bone stress injuries, and osteomyelitis or neoplasm of the distal tibia or calcaneus (5). Soft tissue disorders that may mimic Achilles tendinopathy include isolated retrocalcaneal or superficial calcaneal bursopathy, posterior tibial tendinopathy, or peroneal tendinopathy. Neurologic injuries to consider include S1 radiculopathy, and sural, tibial or Baxter's neuralgia (8). Systemic diseases, including rheumatoid arthritis, sarcoidosis, and spondyloarthropathies, may present with calcaneal enthesopathy (2).

Physical examination begins with inspection. Static foot and ankle alignment and dynamic functional movement should be assessed. Achilles tendinopathy is often associated with either pes cavus or pes planus with hyperpronation. A localized thickening of the Achilles tendon with tenderness is

Table.

Differences in clinical findings, imaging characteristics, and treatment between midportion and insertional Achilles tendinopathy.

Midportion	Insertional
Clinical findings	
Functional impairment, focal tenderness, and localized tendon thickening:	
2 to 6 cm proximal to calcaneal insertion (watershed region)	Calcaneal insertion
More common in general	More common in active patients
Often associated with increase in activity level	
Imaging characteristics	
Increased tendon thickness at affected portion + / – partial tearing + / – neovascularization (US or MRI)	
	Calcaneal enthesophytes/cortical irregularities
	Calcaneal bone marrow edema (MRI)
	Associated bursopathy (ultrasound or MRI)
Treatment	
PT protocol	
Eccentric/Alfredson Eccentric-concentric Heavy-slow resistance	Modified Eccentric/Alfredson (not beyond neutral)
More commonly studied related to injectable treatment options	Surgical treatment often includes retrocalcaneal bursectomy and Haglund's deformity resection

common as fusiform pattern is characteristic of midportion Achilles tendinopathy. Self-reported localized pain, morning stiffness, and palpatory pain are highly sensitive (0.78, 0.88, and 0.84, respectively) (12). Dorsiflexion range of motion may be decreased. Loading of the tendon with single-leg heel raise or hop test and resisted plantar flexion may elicit pain or weakness.

Special tests include the painful arch sign and the Royal London test (12). A positive painful arch sign, often seen in midportion Achilles tendinopathy, occurs when the thickening of the tendon moves with relation to the malleoli during active plantar flexion (12). In a positive Royal London test, tenderness decreases with passive dorsiflexion (12). The combination of tenderness and positive painful arch and Royal London tests provides a sensitivity of 0.59 and specificity of 0.833 for the diagnosis of Achilles tendinopathy in runners (2). In addition, the quality of performing calf raises and number of repetitions may provide information on endurance of muscle and potential targets for rehabilitation.

Imaging

The diagnosis of Achilles tendinopathy is often based on the history and physical examination alone. Imaging, however, may be used to confirm diagnosis and for interventional planning. The severity of pathology on imaging does not always correlate with symptom severity, because abnormal findings may be seen in up to 35% of asymptomatic patients;

therefore, clinical correlation is imperative (5,13,14). In contrast, Achilles tendon pain without imaging findings may be found in up to 19% of symptomatic tendons (5,13,14). In addition, tendon abnormalities on imaging often persist well after functional recovery is achieved; therefore, imaging should not be used to predict symptom resolution or guide return to sport (14).

Lateral weight-bearing radiographs are rarely necessary; however, they can demonstrate soft tissue deviations and calcific or bony abnormalities (2,4). Ultrasound is a cost-effective, dynamic, operator-dependent imaging modality preferred by the authors. It can demonstrate a disruption in the normal fibrillar architecture, tendon hypertrophy, neovascularization on color Doppler, and bursopathy (2,14,15). Ultrasound tissue characterization (UTC) is an advancement in ultrasonography that quantifies tendon structure through a three-dimensional tendon analysis of echotype stability, while ultrasound elastography (UE) uses the physics of vibration to measure the stiffness of soft tissue (5,15,16). While compelling advancements in ultrasound technology, especially for research purposes, UTC and UE have not yet been incorporated into standard of care. Magnetic resonance imaging (MRI) is typically reserved for recalcitrant cases to rule out other soft tissue or bony pathology, further characterize partial tears seen on ultrasound, or to aid in certain interventional or surgical planning. Characteristic MRI and ultrasound images of Achilles tendinopathy are shown in Figure 1.

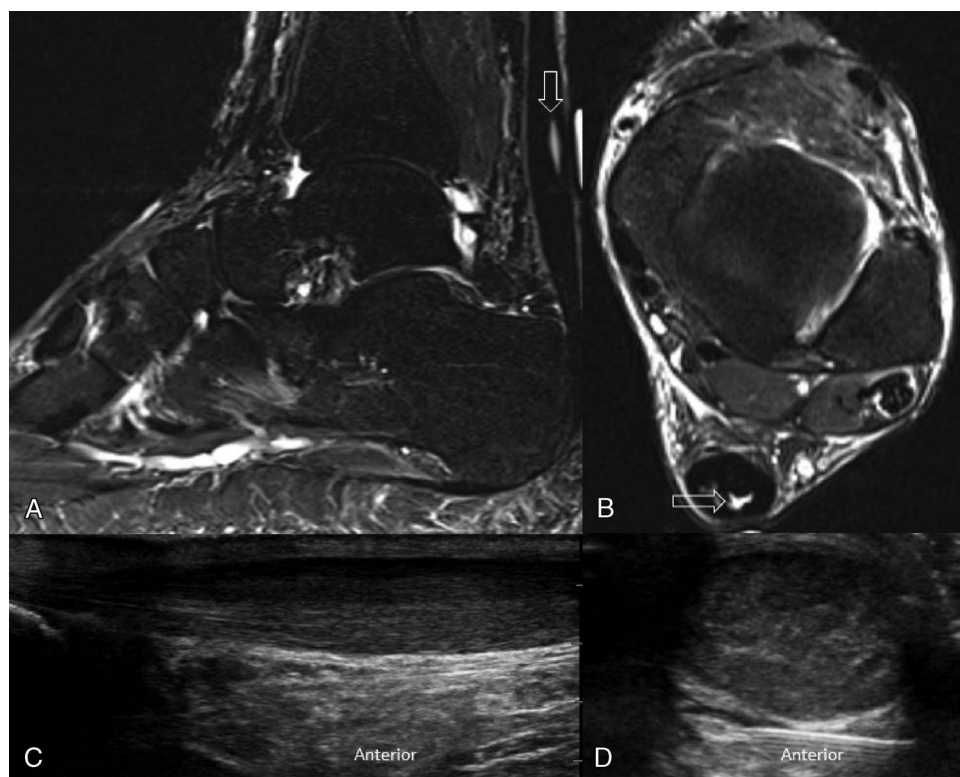


Figure 1: Comparing midportion Achilles tendinopathy on MRI versus ultrasonography. (A) T2 sagittal MRI demonstrating midportion Achilles tendon thickening and intra-substance tear. (B) T2 axial MRI demonstrating midportion Achilles tendon thickening and intra-substance tear. (C) Long axis ultrasound image of Achilles tendon fusiform thickening. (D) Short axis ultrasound image of Achilles tendon thickening. Open arrow indicates intra-substance tearing with increased fluid signal on MRI.

Treatment Strategies

Initial Management

While traditional treatment models of protection, rest, ice, compression, and elevation; activity modification; and patient education may be appropriate early in management of Achilles tendinopathy, **key meta-analyses suggest longer-term benefit of active treatment with eccentric strengthening compared with a wait-and-see approach** (17,18). Accordingly, the mainstay

of treatment for Achilles tendinopathy is PT to restore the tendon's ability to withstand load. Exercise protocols are commonly used with specific differences based on location of tendinopathy.

For midportion tendinopathy, there are six different exercise programs commonly reported: Alfredson protocol, low-volume or “as tolerated” variant of Alfredson protocol, Silbernagel protocol, concentric strength training, heavy slow resistance strength training, and the Stanish protocol (17). The Alfredson protocol is



Figure 2: Comparing the eccentric/Alfredson's protocol for midportion Achilles tendinopathy (A, B, C) with the modified eccentric protocol for insertional Achilles tendinopathy. (A) Two legged plantar flexion position. (B) Shifting weight to involved foot. (C) Eccentric loading of Achilles tendon for midportion tendinopathy. (D) Modified eccentric loading of Achilles tendon only to neutral for insertional tendinopathy.

the most commonly used regimen (13,19). This 12-wk eccentric-loading protocol (Figs. 2A, B, C) incorporates progressive tendon-loading by slowly lowering the affected heel below the edge of a step, eccentrically loading the tendon (13,19). Three sets of 15 repetitions, twice daily, are repeated with the knee in both extension and slight flexion (19). Silbernagel et al. (13,20,21) described a modified eccentric-concentric protocol through four phases: (I) 2 wk of concentric seated and standing heel raises, (II) 4 wk of eccentric single- and double-legged heel raises from the edge of a step, (III) the addition of weights and plyometric exercises through 12 wk, and (IV) the addition of quick rebounding heel raises if necessary. An isolated concentric strength training protocol also has been described (22). In addition, the heavy-slow resistance protocol, described by Beyer et al. (13,23), consists of slow heel raises performed on a seated calf raise or leg press machine, or at a 1.5-in. step, with 3 s for each concentric and eccentric phase. This protocol is performed with progressive repetitions and sets over time, three times per week for 12 wk (13,23). Lastly, the Stanish protocol also focuses on eccentric strength training (24). Overall, Alfredson's eccentric-loading protocol is the most widely used and demonstrates greater improvements in functional measures (VISA-A score) by approximately 20 points at 3 months compared with a wait-and-see approach (17,25). There is a lack of high-level, comparative investigation to prove one program's superiority.

Treatment for insertional Achilles tendinopathy tends to be more challenging (5,26). The modified Alfredson protocol has been described in which the tendon is eccentrically loaded to tibiotalar neutral to avoid posterior ankle impingement (not into dorsiflexion) (Fig. 2A, B, D) (5,26). These exercises are performed in three sets of 15, twice daily for 12 wk (26). Of note, education on the expectation and acceptance of discomfort with loading exercises for both midportion and insertional AT is a necessary component of the rehabilitation progression (13).

In addition to loading protocols for both insertional and midportion tendinopathy, addressing proximal and distal kinetic chain biomechanical abnormalities, incorporation of calf stretches, kinesiology taping, gastroc-soleus trigger point injections, night splints, and proprioceptive neuromuscular facilitation may be appropriate. Manual therapy including osteopathic manipulative treatments and other soft tissue techniques with or without the use of instrument assisted soft tissue mobilization, also may be used as adjunctive tools. The goal of these interventions is symptom modulation; however, limited and conflicting evidence exists regarding their effectiveness. Yet given their low cost and low risk to the patient, they may be considered on an individual basis.

Oral and topical pharmacologic treatments may be used in combination with tendon-loading programs (27). The role of nonsteroidal anti-inflammatory drugs in Achilles tendinopathy has been debated due to the lack of an inflammatory state; however, anti-inflammatories may be useful in the presence of concomitant bursitis or paratenonitis (5,13,27). The use of glyceryl trinitrate (nitroglycerin) transdermal patches may help with tendon blood flow and pain relief through the release of nitric oxide, which is thought to affect collagen production and local blood flow (27). In a randomized control trial (RCT), Paoloni et al. (28) demonstrated improved pain and function with the use of glyceryl trinitrate patches plus

eccentric training for midportion Achilles tendinopathy, when compared with placebo. However, a subsequent RCT by Kane et al. (29) failed to reproduce these clinical benefits. Overall, the authors preferred approach is to work in tandem with PT, optimizing a therapeutic exercise program with an emphasis on eccentric strength training and kinetic chain function. Adjunctive therapeutic options are incorporated on an individualized basis, including a trail of nitroglycerin patches.

Nonoperative Procedural Interventions

For the majority of patients, Achilles tendinopathy will resolve with the aforementioned conservative care. However, nearly one third of cases may go on to develop chronic, recalcitrant symptoms (1). As a result, procedural interventions are frequently used prior to consideration of surgical referral for patients with refractory disease and may include the following: ESWT, high-volume injectate (HVI), prolotherapy, PRP, and other orthobiologic injections, percutaneous tenotomy, and corticosteroid injection (CSI). Two recent systematic reviews with network meta-analysis investigating the efficacy of treatment options for Achilles tendinopathy are important to highlight. Rhim et al. (17) reviewed 22 RCTs and concluded that HVI or ESWT plus eccentric tendon-loading may improve longer-term outcomes (3 to 12 months), while acupuncture demonstrated favorable outcomes in both the short (≤ 3 months) and long term (3 to 12 months). However, van der Vlist et al. (18) reviewed 29 RCTs including 42 different treatment options and concluded that although acupuncture and shockwave therapy with tendon-loading appear superior to a wait-and-see approach at 3 months, no clinically relevant difference exists at 3 to 12 months. While there is promising evidence for many of these treatment options, there are limited studies directly comparing interventions, and therefore, it is difficult to conclude superiority of a given intervention (5,17,18,30). Thus, predictably, there is significant heterogeneity in clinical practice, even among the authors of this article. Therefore, the following discussion is aimed at highlighting more common treatment options rather than providing an exhaustive review.

ESWT is a noninvasive treatment option for refractory Achilles tendinopathy that may stimulate cell-mediated tissue healing and alter maladaptive pain pathways. In a systematic review, Mani-Babu et al. (31) concluded that ESWT is equivalent to a tendon-loading protocol for midportion Achilles tendinopathy in the short term. Numerous systematic reviews suggest that the combined use of ESWT and eccentric-loading may produce superior outcomes than exercise alone in midportion Achilles tendinopathy (17,31). For insertional Achilles tendinopathy, moderate evidence exists to support ESWT as a more effective treatment than eccentric-loading alone (31–33). No evidence-based consensus exists on whether focused or radial shockwave is superior for the treatment of tendinopathy.

HVI with ultrasound guidance targets the interface between the Achilles tendon and the underlying Kager fat pad, designed to mechanically disrupt neovascularization, abnormal nerve sprouting, and/or tissue adhesions, to reduce pain and improve function (27,30). Total injectate volume and composition varies, generally ranging from 30 mL to 50 mL and consisting of saline and local anesthetic, with inclusion of corticosteroid also described in the literature. In the authors' experience,

however, it is best to avoid corticosteroid. A single RCT by Boesen et al. (34) demonstrates improved short- (≤ 3 months) and long-term (> 3 months) functional outcomes with HVI including corticosteroid compared with placebo saline injection, both combined with eccentric loading, for chronic midportion Achilles tendinopathy. Furthermore, tendon scraping, or using the length of the needle to scrape the ventral portion of the tendinopathic region in a longitudinal pattern, is often described with HVI. This technique can be performed with any injectate, as a standalone procedure, or surgically (8).

Prolotherapy involves peritendinous injection of hypertonic dextrose solution theoretically leading to tissue repair through localized mechanical insult and recruitment of cytokines and growth factors (35). A single RCT by Yelland et al. (36) compared eccentric-loading to prolotherapy alone or as combined treatment for midportion Achilles tendinopathy. Those in the prolotherapy groups had earlier improvements in pain and function, but at 12 months, there were no significant differences (36). It may yield good clinical response for insertional Achilles tendinopathy as well, but there is lower level of evidence for this indication (37).

PRP is a concentrated autologous platelet solution potentially contributing to enhanced healing through a complex cell-signaling cascade (38). There is mixed level 1 evidence regarding PRP for midportion Achilles tendinopathy. Older literature demonstrates no benefit compared with placebo, although with methodological concerns limiting interpretation (38), and more recent investigation demonstrates significant benefit compared with placebo at 6 months (34). While lower level of evidence suggests promising use of PRP for Achilles tendinopathy, four recent meta-analyses could not confirm improved efficacy over eccentric training alone (17,30,39,40). Despite this, high patient satisfaction and a favorable safety profile with PRP has been reported in the literature and anecdotally (17,41–43).

Other orthobiologic injectable options include adipose- or bone marrow-derived signaling cells, including stromal vascular fraction (SVF), microfragmented fat, and bone marrow aspirate concentrate (BMAC) (44). There is very limited evidence investigating the use of these injectates, which are more invasive and have higher out-of-pocket costs compared with other treatment options. One RCT has been conducted on Achilles tendinopathy, comparing PRP to SVF in midportion Achilles tendinopathy with earlier improvements in the SVF group, but no significant difference in improvements between groups at 6 months (44). To date, no human studies have investigated BMAC for the treatment of Achilles tendinopathy.

Percutaneous tenotomy is another potential treatment option, which can be performed via needle fenestration of the diseased tendon with or without associated injection, such as PRP, or as a minimally invasive technique of phacoemulsification debridement and aspiration by an ultrasonic vibrating needle attached to a suction and irrigation system (45,46). A retrospective chart review of 34 patients with insertional Achilles tendinopathy who underwent ultrasonic tenotomy found a significant reduction in both short-term (6 and 12 wk) and long-term (1 to 3 years) pain, as well as high patient satisfaction and low complication rate (single superficial skin infection) (45). Other potential complications include the risk of worsening pain, rupture or tears, and deep vein thrombosis (45,46).

While historically used, CSIs are not recommended for the treatment of refractory Achilles tendinopathy due to the non-inflammatory etiology, lack of sustained effect, and concern for potential adverse effects, including tenotoxicity and tendon rupture (2,5,30). While CSI may be considered for concomitant bursitis, postprocedural Achilles tendon rupture has been described in case reports (47). If retrocalcaneal or superficial calcaneal bursa CSI is performed, the procedure should be done with ultrasound guidance to minimize risk and after a thorough discussion with the patient.

Certainly, future high-level comparative studies are necessary to guide evidence-based treatment of refractory Achilles tendinopathy. Furthermore, there are highly variable practice models, including injectate formulation, volume and number of injections, tissue target, image guidance, and postprocedural activity restriction that makes interpretation and comparison of the current literature difficult, as well as variability in out-of-pocket expenses (48). While detailed discussion of these factors is beyond the scope of this review article, all are important to consider during shared decision-making with the patient.

Surgical Treatment

Most cases of Achilles tendinopathy can be treated successfully with nonoperative care. However, surgical treatment should be considered for recalcitrant symptoms. In midportion Achilles tendinopathy, surgical options include a percutaneous longitudinal tenotomy, gastrocnemius recession, tendon debridement, and in severe cases with significant plantar flexion weakness, flexor hallucis longus autograft tendon transfer (2,49). In insertional Achilles tendinopathy, surgical options include Haglunds deformity resection, tendon debridement with reconstruction via suture anchors or autograft reconstruction (2,5). In a recent case series, Alfredson and Spang (50) suggest that a retrocalcaneal bursectomy alone without addressing neighboring tissue, may be an effective treatment option for insertional Achilles tendinopathy. Surgical outcomes are generally positive (5). The most common postoperative complications include wound infection, delayed healing, scar tissue abnormalities, pain, sural nerve injury, and deep vein thrombosis (5).

Return to Sport/Activity Considerations

A gradual, progressive return to recreational activity and sport should be initiated after completion of rehabilitation that is pain free with impact exercise and sports-specific functional activities. **Up to 80% to 90% of those with midportion Achilles tendinopathy are able to return to normal activities with little to no symptoms by 1 year** (13). However, eccentric training may be less effective in sedentary patients (21). Realistic goals need to be set with athletes as recovery may be slow with a high propensity for recurrence (21). **Recurrence rates are more common in those with short recovery periods (< 10 d) and reported in 27% to 44% of soccer players with Achilles tendinopathy** (21,51). The extent of tendon injury, degree of recovery, pain and impairment, and patient demographics, including sport participation, should all be considered when reintroducing activity (21). **An athlete should be monitored closely for symptoms both during and after activity, with lack of symptoms the following day being a helpful determinant for appropriate exercise advancement.**

Prevention

Overall, prevention strategies for Achilles tendinopathy have not been well established within the literature. **Achilles stretching and plantar flexion strengthening protocols are reasonable to institute in impact-sport athletes for both primary and secondary prevention (3). Posterior kinetic chain strengthening beyond the gastrocnemius-soleus complex also may be beneficial. Optimizing both tibiotalar and subtalar joint motion is recommended.** In addition, avoiding fluoroquinolones and other medications that may be associated with tendinopathy is advised, especially in athletes with high impact load (3). Balance training in elite female soccer players also has been suggested as an effective preventative strategy for Achilles tendinopathy (51). Further research is needed to better define the optimal prevention program.

Conclusions

Achilles tendinopathy is a common, degenerative, overuse condition that is diagnosed clinically. Midportion and insertional tendinopathy have distinct features and therapeutic paradigms requiring an individualized approach to treatment. Overall, Achilles tendinopathy has a good clinical prognosis with most patients responding successfully to activity modification and a robust rehabilitation protocol including eccentric tendon loading. However, recalcitrant disease may occur in nearly one third of cases and requires consideration of additional treatments including injections, ESWT, or surgery. There is no standard-of-care for treatment of refractory Achilles tendinopathy. Future research is needed to determine best practice for recalcitrant cases.

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